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MONOLITHIC THREE DIMENSIONAL INTEGRATED CIRCUIT

Abstract

A monolithic three dimensional integrated circuit is provided. The monolithic three dimensional integrated circuit includes a first cell layer having a first cell having a first active component of the monolithic three dimensional integrated circuit. A second layer having a second cell including a second active component. The second cell layer is formed vertically above the first cell layer. The first cell layer having the first active component and the second cell layer having the second active component are formed on a single die. The first cell has a smaller metal pitch than the second cell. A buried via electrically couples the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/245,757 filed Apr. 30, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Historically, elements within an Integrated Circuits (IC) have all been placed in a single two dimensional (2D) active layer with elements interconnected through one or more metal layers that are also within the IC. While such circuits have generally become smaller according to Moore's Law, efforts to miniaturize ICs are reaching their limits in a 2D space and thus, design thoughts have moved to three dimensions. That is, current miniaturization techniques use three-dimensional (3D) integrated circuits (ICs) (3DICs) to achieve higher device packing density, lower interconnect delay, and lower costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion. In addition, the drawings are illustrative as examples of embodiments of the invention and are not intended to be limiting.

[0004] FIG. 1 illustrates an example block diagram of a monolithic three dimensional integrated circuit in accordance with some embodiments.

[0005] FIG. 2 illustrates another example block diagram of a monolithic three dimensional integrated circuit in accordance with some embodiments.

[0006] FIG. 3 illustrates an example three dimensional block diagram of a monolithic three dimensional integrated circuit in accordance with some embodiments.

[0007] FIG. 4 illustrates an example cross-sectional view of the three dimensional block diagram of the monolithic three dimensional integrated circuit of FIG. 3 in accordance with some embodiments.

[0008] FIG. 5 illustrates a process flow for the method for forming a monolithic three dimensional integrated circuit in accordance with some embodiments.

[0009] FIG. 6 is a block diagram illustrating an example of a processing system in accordance with some embodiments.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and

second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0011] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0012] It will be understood that, although the terms “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present invention.

[0013] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it can be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

[0014] Embodiments of the disclosure provide monolithic techniques to form three-dimensional (3D) integrated circuits (ICs) (also referred to as monolithic 3DIC) to miniaturize ICs. For example, the disclosed monolithic 3DICs offer vertical stacking of devices (including logic circuits) on a single die, with the potential to significantly reduce die area and increase die performance. For example, the monolithic 3DICs involve building electronic components and their connections within multiple tiers or multiple layers on a single die. Connections between the multiple tiers within the die are made with buried vias (BVs). Such BVs are much smaller in size than through silicon vias (TSVs) which are used in system in package (SIP) solutions that stack ICs atop one another. Smaller BVs allow 3DICs using BVs to achieve an integration density that is orders of magnitude higher than 3DICs employing TSVs.

[0015] In some embodiments, the layers of monolithic 3DIC are assembled through an ion-cutting process. The use of the layers in this fashion eliminates the need for precise alignment and TSVs. For example, in ion-cutting process, a receptor wafer is prepared with integrated components thereon. An oxide layer forms on a top surface of the receptor wafer. A donor wafer is prepared by subjecting the donor wafer to an ion (typically hydrogen) implantation process. The surface of the donor wafer with the ion implantation is then stacked onto the oxide layer of the receptor wafer. The oxide layer of the receptor wafer bonds with the surface of the donor wafer through an annealing process. The donor wafer is then removed, transferring a silicon layer to the receptor wafer. Additional electronic components and interconnects are fabricated over the transfer silicon layer sequentially. This approach is less expensive than epitaxial growth and eliminates the risk of misalignment, resulting in more functional devices than the techniques that rely on wafer-to-wafer, wafer-to-die, or die-to-die alignment.

[0016] FIG. 1 illustrates an example block diagram of a monolithic 3DIC **100** in accordance with some embodiments. As shown in FIG. 1, 3DIC **100** includes a die **102**. Die **102** includes a plurality of cell layers, for example, a first cell layer **104** and a second cell layer **106**. In some examples, first cell layer **104** and second cell layer **106** have approximately identical horizontal dimensions. In

some other examples, first cell layer **104** is also referred to as first cell tier or a first cell level and second cell layer **106** is also referred to as a second cell tier or a second cell level. Although, monolithic 3DCIC **100** of FIG. **1** is shown to include only two cell layers, that is, first cell layer **104** and second cell layer **106**, it will be apparent to a person with ordinary skill in the art after reading this disclosure that monolithic 3DCIC **100** of FIG. **1** can include more than two cell layers. [0017] As shown in FIG. **1**, die **102** includes a plurality of metal layers. For example, die **102** includes eight metal layers, for example, a zeroth metal layer **M0**, a first metal layer **M1**, a second metal layer **M2**, a third metal layer **M3**, a fourth metal layer **M4**, a fifth metal layer **M5**, a sixth metal layer **M6**, and a seventh metal layer **M7**. As referred to herein, the zeroth metal layer **M0** is generally the lowest metal layer in 3DIC **100**. That is, the zeroth metal layer **M0** is the metal layer closest to a substrate on which the metal layers are formed. The first metal layer **M1** is the metal layer formed above the zero metal layer **M0** without any other metal layer between the zero metal layer **M0** and the first metal layer **M1**. Likewise, the second metal layer **M2** is the next metal layer formed above the first metal layer **M1** without any other metal layer between the first metal layer **M1** and the second metal layer **M2**. The progression of metal layers continues in this fashion until a top metal layer is formed, for example, the seventh metal layer **M7** formed above the sixth metal layer **M6** without any other metal layer between the sixth metal layer **M6** and the seventh metal layer **M7**. It is to be understood that the disclosure is not limited to any specific number of metal layers. In some examples, the zeroth metal layer **M0** is also referred to as a first metal layer **M1** continuing to the seventh metal layer **M7** which is also referred to as an eighth metal layer **M8**. [0018] Continuing with FIG. **1**, first cell layer **104** is located below the zeroth metal layer **M0**. However, in some examples, first cell layer **104** can be located between the zeroth metal layer **M0** and the first metal layer **M1**, between the first metal layer **M1** and the second metal layer **M2**, and so on. First cell layer **104** can include one or more cells. In some examples, the one or more cells of first cell layer **104** can form a logic circuit, a portion of a logic circuit, a logic function, a portion of a logic function, a functional element of a logic circuit, or a portion of a functional element of a logic circuit.

[0019] Second cell layer **106** is placed vertically above first cell layer **104**. For example, second cell layer **106** is placed between the third metal layer **M3** and the fourth metal layer **M4**. However, it will be apparent to a person with ordinary skill in the art that second cell layer **106** can be placed between another pair of metal layers, for example, between the fourth metal layer **M4** and the fifth metal layer **M5**. Second cell layer **106** also includes one or more cells. In some examples, the one or more cells of second cell layer **106** can form a logic circuit, a portion of a logic circuit, a logic function, a portion of a logic function a functional element of a logic circuit, or a portion of a functional element of a logic circuit.

[0020] In some examples, one or more cells of first cell layer **104** and one or more cells of second cell layer **106** together can form a system on chip (SOC). In some other examples, first cell layer **104** and second cell layer **106** of monolithic 3DIC **100** are constructed to support different functionality and comply with differing design criteria. For example, monolithic 3DIC **100** can have one or more analog layers, digital layers, layers with higher voltage threshold, layers with lower leakage current, power supply layers, layers of different material to implement components that need different base materials and the like. In some examples, and unlike the stacked dies of system in package (SIP) arrangements, the upper layers of monolithic 3DIC **100**, that is, second cell layer **106**, may be the same size as the lower layers, that is, first cell layer **104** because no external wiring connections are required. By having the myriad layers within a single 3DIC **100**, an entire system may be provided in a single IC and thus provide a system on chip (SOC).

[0021] In some examples, one or more cells in first cell layer **104** can be connected with one or more cells of second cell layer **106** using one or more buried vias (BVs). FIG. **2** illustrates another example block diagram of monolithic 3DIC **100** in accordance with some embodiments. As shown in FIG. **2**, monolithic 3DIC **100** a first cell **202**, a second cell **204**, and a buried via (BV) **206**. In

examples, first cell **202** is formed in first cell layer **104** and second cell **204** is formed vertically above first cell **202** in second cell layer **106**. First cell **202** is electrically coupled with second cell **204** through BV **206**. In some examples, BV **206** is much smaller in size than the previously described TSVs, thus allowing monolithic 3DIC **100** using BV **206** to achieve an integration density that is orders of magnitude higher than 3DICs employing TSVs.

[0022] In some examples, first cell **202** includes a first active component and second cell **204** includes a second active component. Each of the first active component and the second active component can include an entire device or portions of the entire device, such as a transistor, diode, capacitor, resistor, or inductor, or can include a group of several devices arranged to achieve some particular function, such as an inverter, a flip-flop, a memory cell, or an amplifier, among others.

[0023] In some examples, first cell **202** has a lower current leakage than second cell **204**. In other examples, first cell **202** is optimized for a higher speed than second cell **204**. In some other examples, first cell **202** and second cell **204** when electrically coupled form an integrated circuit or a functional circuit. The functional circuit can be at least one of: computation, digital processing, analog processing, radio frequency (RF) signal processing, analog/mixed signal processing, power management, sensor, power supply, battery, memory, digital logic, low leakage, low noise/high gain, clock, combinatorial logic, and sequential logic.

[0024] In some examples, first cell **202** is formed in first portion, that is, first four layers (that is, M0 to M3) of the eight metal layers of die **102** and second cell **204** is formed in the remaining four layers (that is, M4 to M7) of the eight metal layers of die **102**. In other examples, first cell **202** is formed in zeroth metal layer M0 die **102** and second cell **204** is formed in the fourth metal layer M4 of the eight metal layers of die **102**. In such examples, local connectors of first cell **202** are formed in the first metal layer M1 of die **102** and global connectors of first cell **202** are formed in the second metal layer M2 and third metal layer M3. Similarly, local connectors for second cell **204** are formed in the fifth metal layer M5. In addition, global connectors for second cell **204** can be formed in the sixth metal layer M6 and the seventh metal layer M7.

[0025] In example embodiments, second cell **204** of monolithic 3DIC **100** is connected to first cell **202** through BV **206**. For example, and as shown in FIG. 2, BV **206** connects a metal plate of first cell **202** formed in the third metal layer M3 with a metal plate of second cell **204** formed in the fifth metal layer M5. However, it will be apparent to a person with ordinary skill in the art after reading this disclosure that BV **206** can connect first cell **202** with second cell **206** by connecting the metal plates formed in other metal layers. For example, BV **206** can connect first cell **202** with second cell **206** by connecting a metal plate of second cell **204** formed in one of the fourth metal layer M4 or the fifth metal layer M5 with a metal plate of first cell **202** formed in one of the first metal layer M1, the second metal layer M2, or the third metal layer M3. In some examples, a metal type, a width, a height, and a thickness of BV **206** can depend on a cell type of each of first cell **202** and second cell **204** and may be different for a different type of cells.

[0026] In some examples, first cell layer **104** includes higher cost cells and second layer **106** includes lower cost cells. For example, a pitch of first cell **202** of first cell layer **104** is smaller than the pitch of second cell **204** of second cell layer **106**. In addition, second cell layer **106** includes a fewer number of cells than first cell layer **104**. In some examples, a ratio of number of cells in first cell layer **104** and second cell layer **106** can be predetermined.

[0027] FIG. 3 illustrates an example three dimensional block diagram of monolithic 3DIC **100** in accordance with some embodiments. As shown in FIG. 3, monolithic 3DIC **100** includes first substrate plates **302**, zeroth metal plates **304**, first metal plates **306**, second metal plates **308**, third metal plates **310**, second substrate plates **312**, fourth metal plates **314**, fifth metal plates **316**, and BV **318**. In some examples, first substrate plates **302**, zeroth metal plates **304**, first metal plates **306**, second metal plates **308**, and third metal plates **310** can constitute first cell layer **104** or first cell **202**. In some other examples, second substrate plates **312**, fourth metal plates **314**, and fifth metal plates **316** can constitute second cell layer **106** or second cell **204**.

[0028] In some examples, and as shown in FIG. 3, first substrate plates **302** are formed below the zeroth metal layer **M0** and zeroth metal plates **304** are formed in the zeroth metal layer **M0**. Moreover, first metal plates **306** are formed in the first metal layer **M1** and second metal plates **308** are formed in the second metal layer **M2**. Furthermore, third metal plates **310** are formed in the third metal layer **M3**. In addition, second substrate plates **312** are formed between the third metal layer **M3** and the fourth metal layer **M4**. Moreover, fourth metal plates **314** are formed in the fourth metal layer **M4** and fifth metal plates **316** are formed in the fifth metal layer **M5**.

[0029] In some examples, first substrate plates **302** include a larger number of unit plates, and hence, a larger number of cells, than second substrate plates **312**. For example, and as shown in FIG. 3, first substrate plates **302** include twelve unit plates (labeled as, **302a**, **302b**, **302c**, **302d**, **302e**, **302f**, **302g**, **302h**, **302i**, **302j**, **302k**, and **302l**) while second substrate plates **312** include eight unit plates (labeled as, **312a**, **312b**, **312c**, **312d**, **312e**, **312f**, **312g**, and **312h**). However, it will be apparent to a person with ordinary skill in the art after reading this disclosure that a different number of unit plates are foreseeable for both first substrate plates **302** and second substrate plates **312**. In addition, although zeroth metal plates **304** are shown to be formed in unit plate **302h**, it will be apparent to a person with ordinary skill in the art after reading this disclosure that zeroth metal plates **304** can be formed any of unit plates labeled as **302a**, **302b**, **302c**, **302d**, **302e**, **302f**, **302g**, **302h**, **302i**, **302j**, **302k**, and **302l**. Moreover, although fourth metal plates **314** are shown to be formed in unit plate **312d**, it will be apparent to a person with ordinary skill in the art after reading this disclosure that fourth metal plates **314** can be formed any of unit plates labeled as **312a**, **312b**, **312c**, **312d**, **312e**, **312f**, **312g**, and **312h**.

[0030] In some examples, zeroth metal plates **304** have a shorter, a smaller pitch, or a denser pitch, and hence, a higher cost, than fourth metal plates **314**. For example, and as shown in FIG. 3, zeroth metal plates **304** include five metal plates in an unit plate while fourth metal plates **314** include three metal plates in an unit plate. However, it will be apparent to a person with ordinary skill in the art after reading this disclosure that a different number of metal plates are possible in a unit plate for both zeroth metal plates **304** and fourth metal plates **314**.

[0031] In examples, BV **318** connects fifth metal plates **316** with third metal plates **318**. That is, BV **318** connects fifth metal plates **316** of second cell **204** with third metal plates **318** of first cell **202** thereby electrically coupling first cell **202** with second cell **204**. In some examples, first metal plates **306** are also referred to as local routes for first cell **202** and can be used to connect one cell of one cell layer with another cell of the same cell layer. For example, the local routes are used for electrically coupling first cell **202** with another cell in first cell layer **104**. In some other examples, second metal plates **308** and third metal plates **310** are also referred to as global routes of first cell **202** and can be used to connect one cell of one cell layer with another cell of another cell layer. For example, the global routes are used for electrically coupling first cell **202** second cell **204** of second cell layer **106**. In example embodiments, BV **318** is separated from second substrate plates **312** by a predetermined distance. In examples, BV **318** is similar to or substantially same as BV **206** of FIG. 2.

[0032] FIG. 4 illustrates an example cross-sectional view of the three dimensional block diagram of monolithic 3DIC **100** of FIG. 3 in accordance with some embodiments. As shown in FIG. 4, an edge of side of BV **318** is at least “d” distance apart from a closest edge or side of second substrate plates **312** (for example, unit plate **312f**). The “d” distance can be predetermined and may depend on a type of cells being connected by BV **118**, types of materials used for forming each of BV **118**, second substrate plates **112**, fifth metal plates **316**, and third metal plates **310**.

[0033] FIG. 5 is a flowchart of a method **500** for forming monolithic three dimensional integrated circuit in accordance with some embodiments. For example, method **500** can be performed to form monolithic 3DIC **100** described above with reference to FIGS. 1-4 of the disclosure. The method **500** can be performed, for example, by a computer **600** as discussed with reference FIG. 6. In some embodiments, method **500** can be stored as instructions on a non-transitory computer readable

medium which can be executed by a processor to perform method **500**.

[0034] At block **510** of method **500**, a first cell layer having a first cell is formed. For example, first cell layer **104** having first cell **202** is formed. In some examples, first cell **202** is formed in a first two metal layers of die **102** having eight metal layers. In some other examples, first cell **202** is formed in a first four metal layers of die **102** having eight metal layers.

[0035] At block **520** of method **500**, a second cell layer having a second cell is formed. For example, second cell layer **106** having second cell **204** is formed. In some examples, second cell **204** is formed in a next two metal layers of die **102**. That is, second cell **204** is formed in the fifth metal layers **M4** and the sixth metal layer **M5** of die **102**. Second cell layer **106** is formed vertically above first cell layer **104**.

[0036] At block **530** of method **500**, the first cell of the first cell layer is electrically coupled with the second cell of the second cell layer through a buried via. For example, first cell **202** is electrically coupled to second cell **204** via **BV 206**. In some examples, first cell **202** and second cell **204** when electrically coupled can form monolithic 3DIC **100**.

[0037] FIG. **6** is a block diagram illustrating an example of a processing system **600** in accordance with some embodiments disclosed herein. Processing system **600** may be used to form monolithic 3DIC **100** in accordance with various processes discussed herein. Processing system **600** includes a processing unit **610**, such as a desktop computer, a workstation, a laptop computer, a dedicated unit customized for a particular application, a smart phone or tablet, etc. Processing system **600** may be equipped with a display **614** and one or more input/output devices **612**, such as a mouse, a keyboard, touchscreen, printer, etc. Processing unit **610** also includes a processing unit (PU) **620**, storage device **622**, a mass storage device **624**, a video adapter **626**, and an I/O interface **628** connected to a bus **630**.

[0038] Bus **630** may be one or more of any type of several bus architectures including a memory bus or memory controller, a peripheral bus, or video bus. PU **620** may comprise any type of electronic data processor, and storage device **622** may comprise any type of system memory, such as static random access memory (SRAM), dynamic random access memory (DRAM), or read-only memory (ROM).

[0039] Mass storage device **624** may comprise any type of storage device configured to store data, programs, and other information and to make the data, programs, and other information accessible via bus **630**. Mass storage device **624** may comprise, for example, one or more of a hard disk drive, a magnetic disk drive, an optical disk drive, flash memory, or the like.

[0040] The term computer readable media as used herein may include computer storage media such as the system memory and storage devices mentioned above. Computer storage media may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, or program modules. Storage device **622** and mass storage device **624** are computer storage media examples (e.g., memory storage).

[0041] Computer storage media may include RAM, ROM, electrically erasable read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other article of manufacture which can be used to store information and which can be accessed by processing device **600**. Any such computer storage media may be part of processing device **600**. Computer storage media does not include a carrier wave or other propagated or modulated data signal.

[0042] Communication media may be embodied by computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” may describe a signal that has one or more characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media may

include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, radio frequency (RF), infrared, and other wireless media.

[0043] Video adapter **626** and I/O interface **628** provide interfaces to couple external input and output devices to processing unit **610**. As illustrated in FIG. **6**, examples of input and output devices include display **614** coupled to video adapter **626** and I/O device **612**, such as a mouse, keyboard, printer, and the like, coupled to I/O interface **628**. Other devices may be coupled to processing unit **610**, and additional or fewer interface cards may be utilized. For example, a serial interface card (not shown) may be used to provide a serial interface for a printer. Processing unit **610** also may include a network interface **640** that may be a wired link to a local area network (LAN) or a wide area network (WAN) **616** and/or a wireless link.

[0044] Embodiments of processing system **600** may include other components. For example, processing system **600** may include power supplies, cables, a motherboard, removable storage media, cases, and the like. These other components, although not shown, are considered part of processing system **600**.

[0045] In some examples, instructions or software code is executed by PU **620** to perform refresh operations. The instructions or the software code may be accessed by PU **620** via bus **630** from storage device **622**, mass storage device **624**, or the like, or remotely through network interface **640**. Further, in some examples, the refresh operations instructions may be received through I/O interface **628** and/or stored in storage device **622** or mass storage device **624** in accordance with various methods and processes implemented by the software code.

[0046] Disclosed embodiments thus provide a monolithic three dimensional integrated circuit comprising: a first cell layer comprising a first cell having a first active component of a monolithic three dimensional integrated circuit; a second cell layer comprising a second cell having a second active component, wherein the second cell layer is formed vertically above the first cell layer, wherein the first cell layer having the first active component and the second cell layer having the second active component are formed on a single die, and wherein the first cell has a shorter metal pitch than the second cell; and a buried via electrically coupling the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer.

[0047] In accordance with other disclosed examples a monolithic three dimensional integrated circuit comprising: a die comprising a plurality of metal layers; a first cell layer formed below a first metal layer of the plurality of metal layers of the die, the first cell layer comprising a first cell having a first active component; a second cell layer formed between a fourth metal layer and a fifth metal layer of the plurality of layers of the die, wherein the second cell layer comprises a second cell having a second active component, wherein the second cell layer is formed vertically above the first cell layer, and wherein the first cell has a shorter metal pitch than the second cell; and a buried via electrically coupling the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer.

[0048] In accordance with still further disclosed examples, a method of forming a monolithic three dimensional integrated circuit comprises: forming a first cell layer comprising a first cell having a first active component; forming a second cell layer comprising a second cell having a second active component, wherein forming the second cell layer comprises forming the second cell layer vertically above the first cell layer, and wherein the first cell layer having the first active component and the second cell layer having the second active component are formed on a single die, and wherein the first cell has a shorter metal pitch than the second cell; and electrically coupling the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer through a buried via.

[0049] This disclosure outlines various embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and

structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A monolithic three-dimensional integrated circuit comprising: a first cell layer formed below a zeroth metal layer of the plurality of metal layers of a single die, the first cell layer comprising a first cell having a first active component of the monolithic three-dimensional integrated circuit; a second cell layer formed in between a third metal layer and a fourth metal layer of the plurality of metal layers of the single die, the second cell layer comprising a second cell having a second active component, wherein the second cell layer is formed vertically above the first cell layer, wherein the first cell has a smaller metal pitch than the second cell, and wherein the plurality of metal layers are disposed in the vertical direction; and a buried via electrically coupling the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer, wherein the buried via connects the first cell with the second cell through a third metal layer and a fifth metal layer of the plurality of metal layers of the single die.
2. A method of forming a monolithic three-dimensional integrated circuit, comprising: forming a first cell layer below a first metal layer of a plurality of metal layers of a single die, the first cell layer comprising a first cell having a first active component; forming a second cell layer between a fourth metal layer and a fifth metal layer of the plurality of metal layers of the single die, wherein the second cell layer comprises a second cell having a second active component, wherein the second cell layer is formed vertically above the first cell layer, wherein the second cell layer and the first cell layer have identical horizontal dimensions; and forming a buried via that electrically couples the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer, wherein the buried via electrically couples the first cell with the second cell through a fourth metal layer and a sixth metal layer of the plurality of metal layers of the single die.
3. A monolithic three-dimensional integrated circuit, comprising: a first cell layer formed below a zeroth metal layer of a plurality of metal layers of a single die, the first cell layer comprising a first cell having a first active component; a second cell layer formed in between a third metal layer and a fourth metal layer of the plurality of metal layers of the single die, the second cell layer comprising a second cell having a second active component, wherein the second cell layer is formed vertically above the first cell layer, and wherein both the first cell layer having the first active component and the second cell layer having the second active component are formed in the single die comprising the plurality of metal layers disposed in a vertical direction; and a buried via electrically coupling the first active component of the first cell of the first cell layer with the second active component of the second cell of the second cell layer, wherein the buried via connects the first cell with the second cell through a third metal layer and a fifth metal layer of the plurality of metal layers of the single die.
4. The monolithic three-dimensional integrated circuit of claim 3, wherein the first cell layer has a greater number of cells than the second cell layer.
5. The monolithic three-dimensional integrated circuit of claim 3, wherein the buried via is separated by a predetermined distance from a nearest edge of a substrate of the second cell.
6. The monolithic three-dimensional integrated circuit of claim 3, further comprising local routes for the first cell, wherein the local routes are formed in a first metal layer of the plurality of metal layers.
7. The monolithic three-dimensional integrated circuit of claim 3, further comprising global routes

- for the first cell, wherein the global routes connect the first cell with the second cell, and wherein the global routes for the first cell are formed in a second metal layer and the third metal layer of the plurality of metal layers.
- 8.** The monolithic three-dimensional integrated circuit of claim 3, wherein the first cell has a lower leakage current than the second cell.
- 9.** The monolithic three-dimensional integrated circuit of claim 3, wherein the buried via is smaller in size than a through silicon via.
- 10.** The monolithic three-dimensional integrated circuit of claim 1, wherein the first cell layer has a greater number of cells than the second cell layer.
- 11.** The monolithic three-dimensional integrated circuit of claim 1, wherein the buried via is separated by a predetermined distance from a nearest edge of a substrate of the second cell.
- 12.** The monolithic three-dimensional integrated circuit of claim 1, further comprising local routes for the first cell, wherein the local routes are formed in a first metal layer of the plurality of metal layers.
- 13.** The monolithic three-dimensional integrated circuit of claim 1, further comprising global routes for the first cell, wherein the global routes connect the first cell with the second cell, and wherein the global routes for the first cell are formed in a second metal layer and the third metal layer of the plurality of metal layers.
- 14.** The monolithic three-dimensional integrated circuit of claim 1, wherein the first cell has a lower leakage current than the second cell.
- 15.** The monolithic three-dimensional integrated circuit of claim 1, wherein the buried via is smaller in size than a through silicon via.
- 16.** The method of claim 2, wherein the first cell layer has a greater number of cells than the second cell layer.
- 17.** The method of claim 2, wherein the buried via is separated by a predetermined distance from a nearest edge of a substrate of the second cell.
- 18.** The method of claim 2, further comprising local routes for the first cell, wherein the local routes are formed in a first metal layer of the plurality of metal layers.
- 19.** The method of claim 2, further comprising global routes for the first cell, wherein the global routes connect the first cell with the second cell, and wherein the global routes for the first cell are formed in a second metal layer and the third metal layer of the plurality of metal layers.
- 20.** The method of claim 2, wherein the first cell has a lower leakage current than the second cell.
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